

Pramo Jewels

Migration Strategy

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Version: 1.0

Status: Approved for Development

Field	Value
Project	Pramo Jewels Premium E-Commerce Platform
Purpose	Define standards for database schema evolution, data migration and release sa
Audience	Backend Developers, DBAs, DevOps Engineers, QA
Version	1.0

1. Migration Objectives

Ensure database changes are version-controlled, repeatable, reversible where practical and safe across development, staging and production environments.

2. Migration Workflow

Design Change → Review → Laravel Migration → Local Testing → CI Validation → Staging Deployment → Backup → Production Deployment → Verification

3. Migration Standards

- One logical change per migration.
- Use descriptive migration names.
- Define explicit foreign keys and indexes.
- Avoid destructive schema changes in a single deployment.

4. Data Migration

Use Laravel seeders and dedicated migration scripts for initial data, reference data and legacy data imports. Validate migrated records before production cutover.

5. Rollback Strategy

Implement down() methods whenever feasible. For irreversible operations, create compensating migrations and restore from verified backups if required.

6. Zero-Downtime Practices

- Add nullable columns before enforcing constraints.
- Backfill data in batches.
- Deploy code compatible with both old and new schemas during transition.
- Remove deprecated columns only after successful adoption.

7. Backup & Recovery

Take a verified database backup before every production migration. Test restoration procedures regularly and retain backups according to policy.

8. Validation & Testing

Run automated migration tests, schema validation, foreign key checks, application smoke tests and performance verification after deployment.

9. Governance

All migrations require code review, CI approval and release documentation. Emergency migrations must be logged and retrospectively reviewed.